

Q5

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```
library(readxl)

### Import CSV file
data1 <- read.csv("/Users/erinanderson/Desktop/R_Work/Stats_HW_1/BMI.csv")

head(data1)

##   X Weight.kg.      BMI
## 1 1          110 32.14025
## 2 2          104 27.35043
## 3 3          103 36.06316
## 4 4           80 31.64432
## 5 5          101 27.39800
## 6 6          107 45.70892

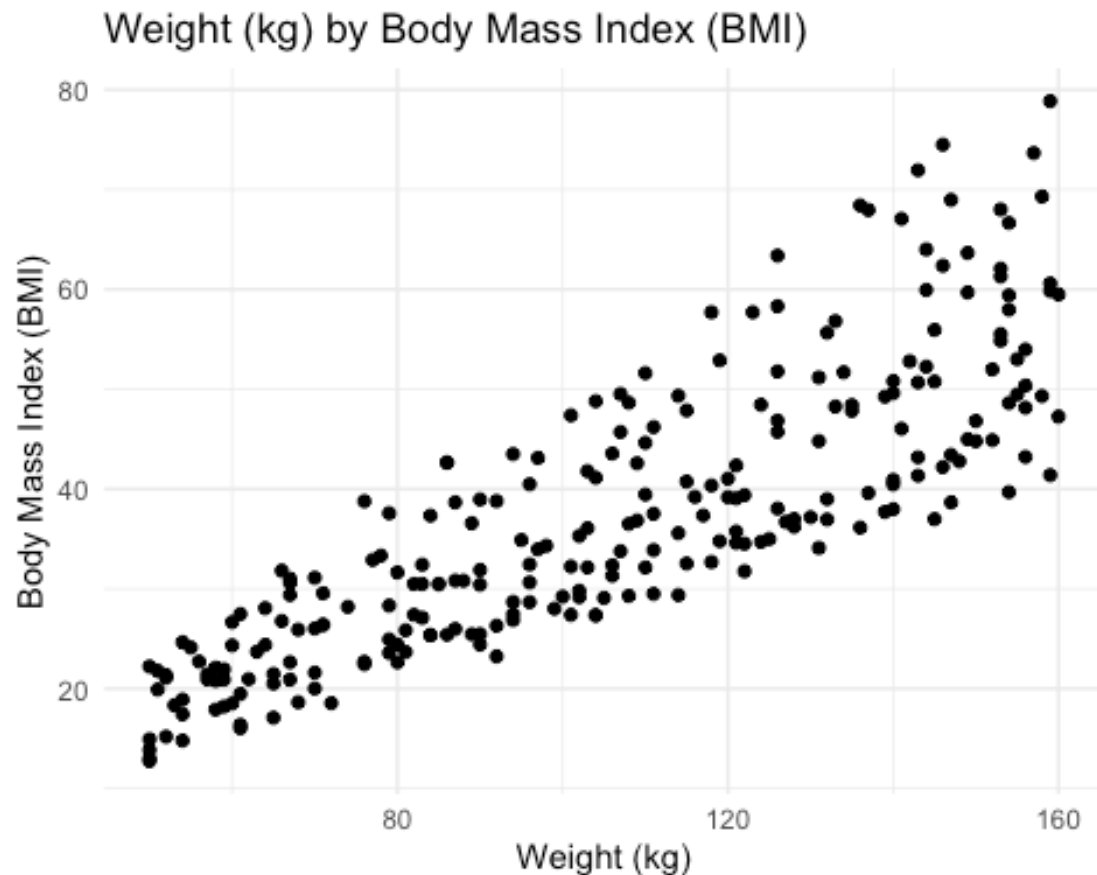
library(ggplot2)

## Warning: package 'ggplot2' was built under R version 4.4.3

### Making the scatter plot and naming the x, y, and title

p <- ggplot(data1, aes(x = Weight.kg., y = BMI)) +
  geom_point() +
  labs(
    x = "Weight (kg)",
    y = "Body Mass Index (BMI)",
    title = "Weight (kg) by Body Mass Index (BMI)") +
  theme_minimal()

print(p)
```



Hand-calculation of the model coefficients and regression equation
 #### using R as a calculator.

Calculate the mean of x and y

```
mean_x <- mean(data1$Weight.kg.); mean_y <- mean(data1$BMI)
print(cbind(mean_x, mean_y))
```

```
##          mean_x  mean_y
## [1,] 105.698  37.3941
```

Calculate the variance and covariance of x and y

```
var_x <- var(data1$Weight.kg.); var_y <- var(data1$BMI); cov_xy <-
cov(data1$Weight.kg., data1$BMI)
print(cbind(var_x, var_y, cov_xy))
```

```
##          var_x    var_y    cov_xy
## [1,] 1086.495  192.4161  383.6014
```

Calculate the sum of squares (ss) of x, y, and covariance of xy

```
n <- length(data1$BMI)
SS_xx <- (n-1)*var_x
SS_xy <- (n-1)*cov_xy
SS_yy <- (n-1)*var_y
print(cbind(SS_xx, SS_xy, SS_yy))
```

```
##          SS_xx    SS_xy    SS_yy
## [1,] 275969.7 97434.75 48873.7

### Calculate the model parameters including slope (b1), intercept (b0),
regression line (yhat), and standard error (e)
b1 <- SS_xy/SS_xx
b0 <- mean_y - b1*mean_x
yhat <- b0 + b1*(data1$Weight.kg.)
e <- (data1$BMI)-yhat
head(cbind(b0, b1))

##          b0          b1
## [1,] 0.07602281 0.3530631

### Calculate sum of square errors (SSE), mean square error (MSE), and root
MSE (s)
SSE <- sum(e^2)
MSE <- SSE/(n-2)
s <- sqrt(MSE)

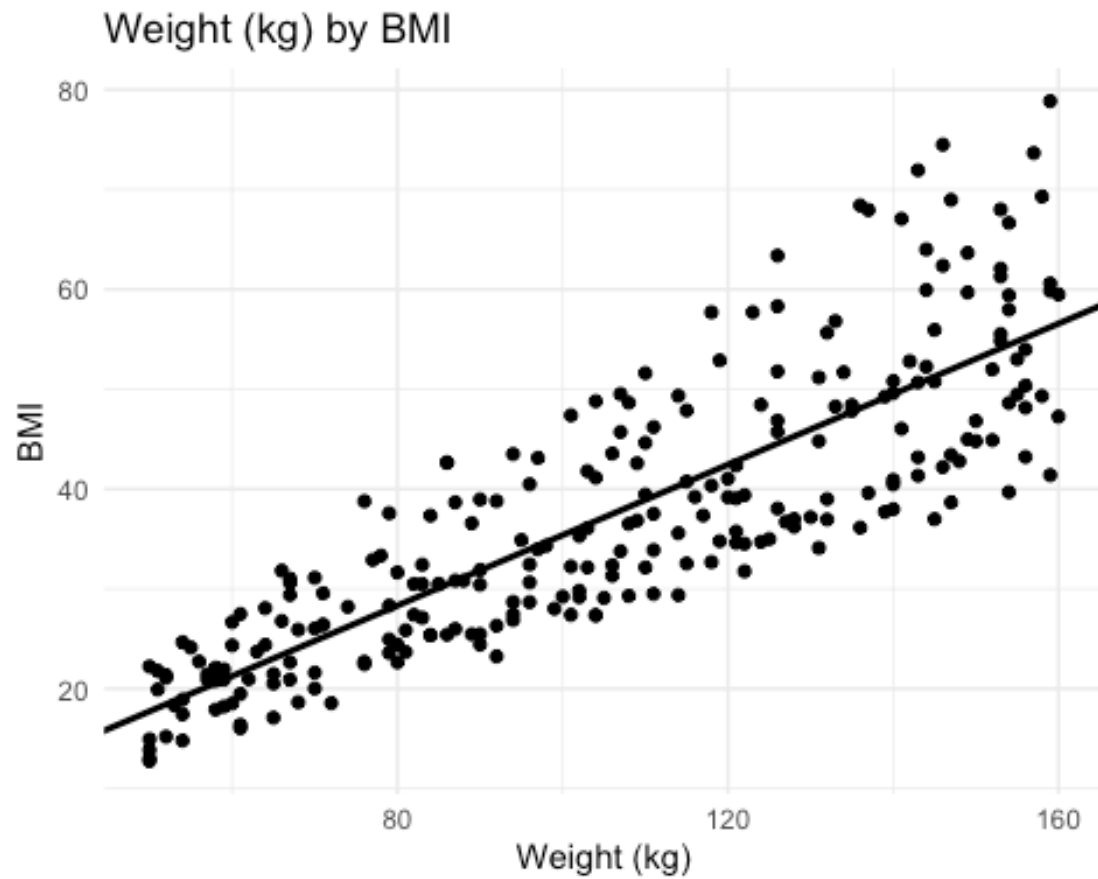
print(cbind(SSE, MSE, s))

##          SSE          MSE          s
## [1,] 14473.08 57.20585 7.563455

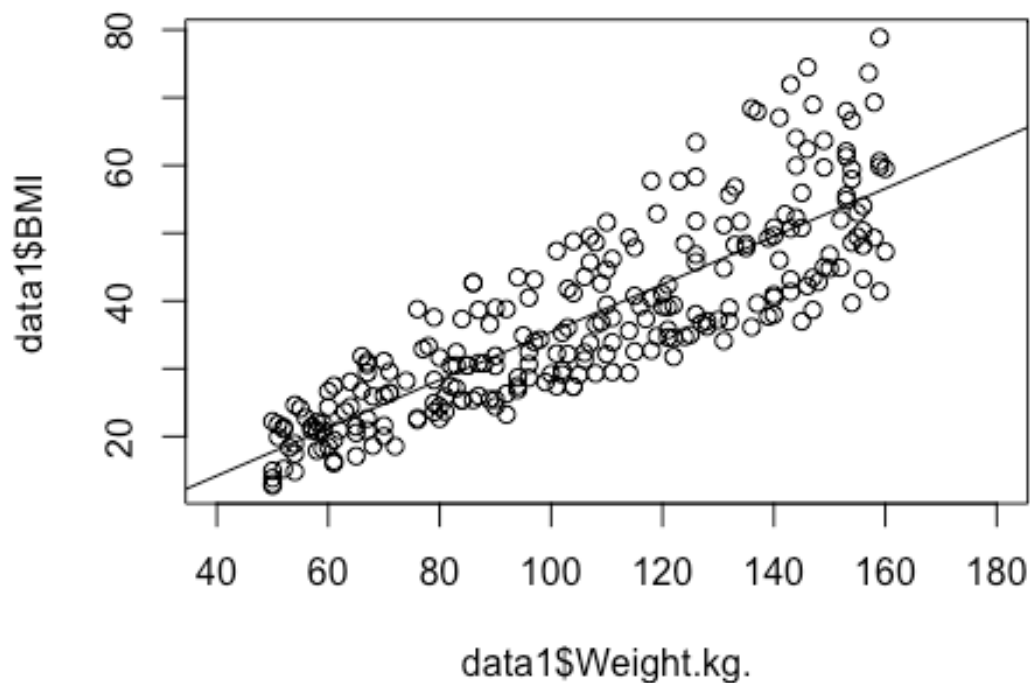
library(ggplot2)

### Adding the linear regression to the original scatter plot using the
calculated slope (b1) and intercept (b0) from above
q <- ggplot(data1, aes(x = Weight.kg., y = BMI)) +
  geom_point() +
  geom_abline(
    intercept = b0,
    slope = b1,
    linewidth = 1) +
  labs(
    x = "Weight (kg)",
    y = "BMI",
    title = "Weight (kg) by BMI") +
  theme_minimal()

print(q)
```



```
### Doubling checking my method against base R (I am less familiar with base R)  
plot(data1$Weight.kg.,data1$BMI,xlim=c(40,180))  
abline(a=b0,b=b1)
```



```
#Re-running the regression analysis except using lm() on the data
lm1 <- lm(BMI ~ Weight.kg., data1)
summary(lm1)
```

```
##
## Call:
## lm(formula = BMI ~ Weight.kg., data = data1)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -14.8241  -5.6286  -0.5016   4.6481  22.8666
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   0.07602    1.59380   0.048   0.962
## Weight.kg.    0.35306    0.01440  24.522 <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 7.563 on 253 degrees of freedom
## Multiple R-squared:  0.7039, Adjusted R-squared:  0.7027
## F-statistic: 601.3 on 1 and 253 DF,  p-value: < 2.2e-16
```

The slope (b_1) and intercept (b_0) of the model match what was hand calculated. The slope = 0.353 (kg/BMI units) and the intercept = 0.076 (BMI units). This equation is modeling the mean change in BMI for a person with a weight between 50kg and 160kg. In this example the intercept is not interpretable because it crosses the y axis at $X = 0$ (a person weighing 0kg). The data set does not include any provided data at that weight and so we cannot use the intercept to interpret the mean BMI of a 0kg person.

Confidence Interval for the Slope
`confint(lm1, 'Weight.kg.', level=0.95)`

```
##                2.5 %    97.5 %  
## Weight.kg. 0.3247088 0.3814175
```

$H_0: b_1 = 0$; $H_a: b_1 \neq 0$.

T-test

```
t_test <- b1/MSE  
t_test
```

```
## [1] 0.006171801
```

Compare to t-test table at 253 df = 1.960 for a two tailed t-test with $\alpha = 0.05$. The critical value is greater than the test statistic so we can reject the null. There is a significant mean change in BMI per kg.